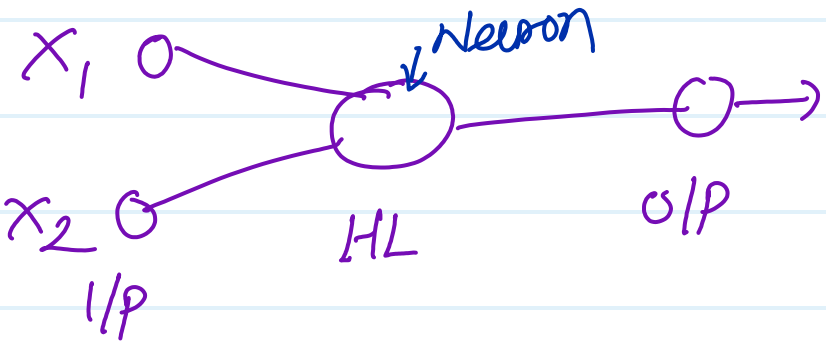
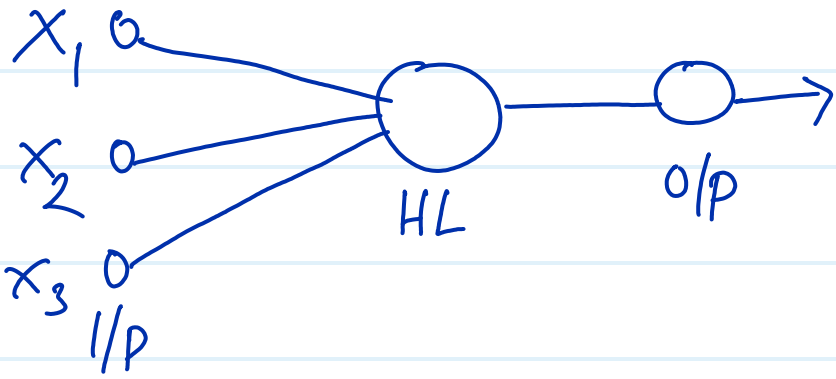
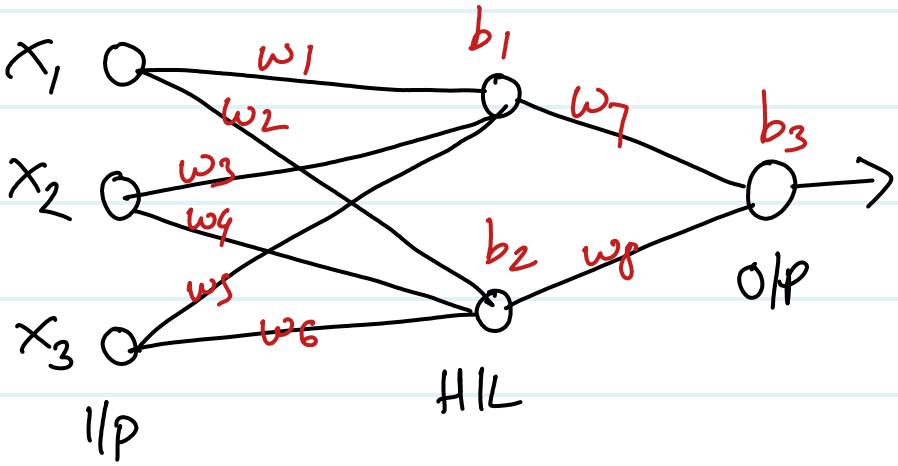
1) Neuron :- single unit



2) Perceptron :-



3) weight & bias

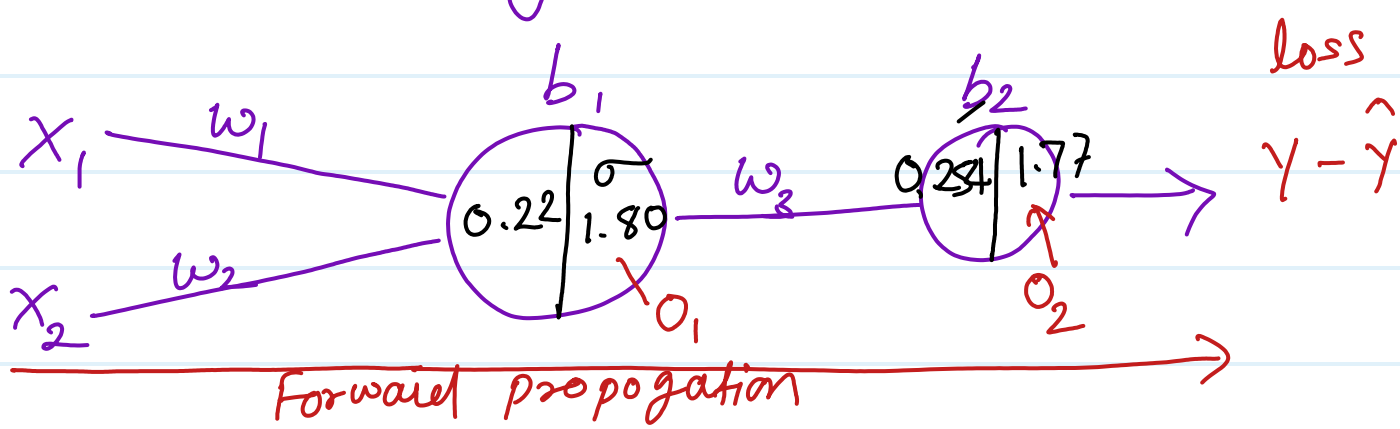


$$z = wx + b \quad \text{simple}$$

$$z = x_1 w_1 + x_2 w_3 + x_3 w_5 + b_1$$

$$z = x_1 w_2 + x_2 w_4 + x_3 w_6 + b_2$$

4) Forward propogation:-



x_1	x_2	y
2	5	1

assumed $w_1 = 0.01$ $\quad b_1 = 0.1$

$w_2 = 0.02$

$w_3 = 0.03$ $\quad b_2 = 0.2$

$$z = x_1 w_1 + x_2 w_2 + b_1$$

$$\Rightarrow 2 \times 0.01 + 5 \times 0.02 + 0.1$$

$$\Rightarrow 0.22$$

Activate function

$$\sigma = \frac{1}{1+e^{-2}}$$

$$= \frac{1}{1+e^{-(0.22)}}$$

$$\Rightarrow 0.554$$

$$Z_1 = O_1 w_3 + b_2$$

$$= 0.554 \times 0.03 + 0.2$$

$$\Rightarrow 0.21$$

$$\sigma = \frac{1}{1+e^{-2}} = \frac{1}{1+e^{-(0.21)}}$$

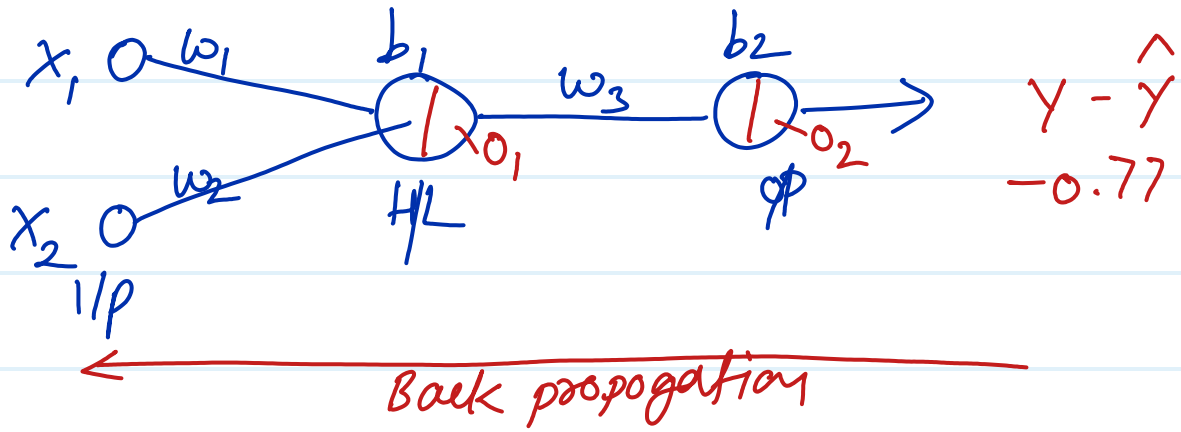
$$\Rightarrow 0.552$$

loss $\quad Y = 1 \quad , \quad \hat{Y} = 0.552$

$$\text{loss} = Y - \hat{Y} = 1 - 0.552$$

$$= 0.45$$

5) Back propogation :-



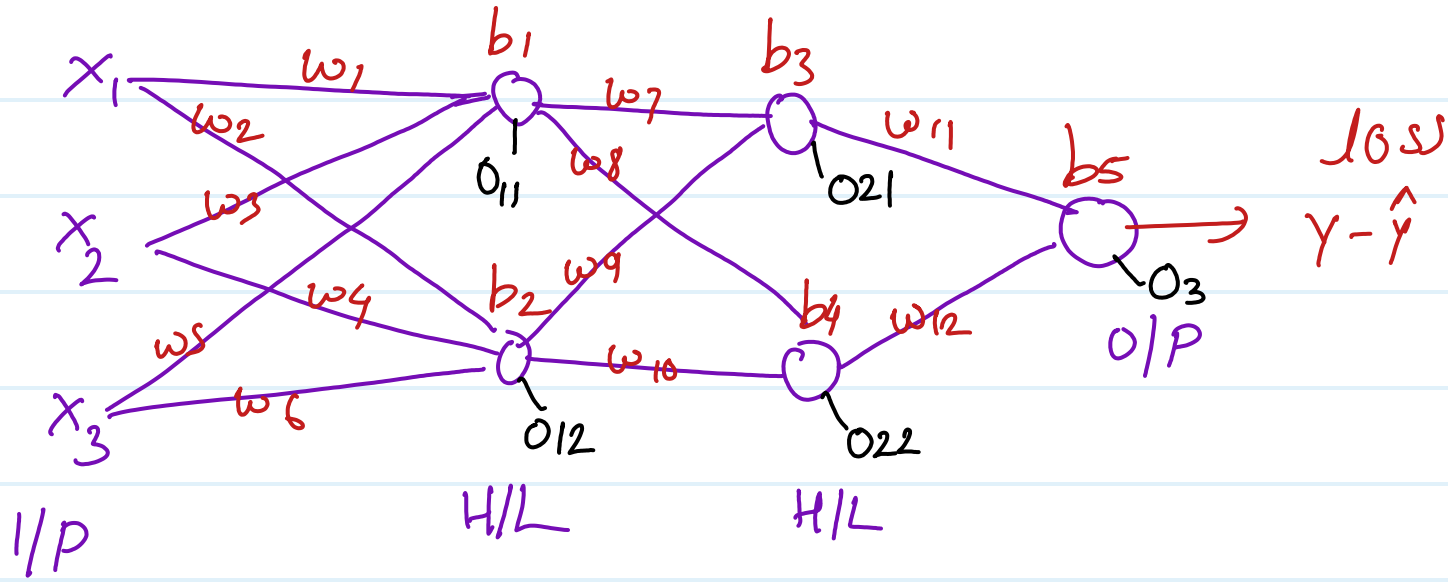
Optimizer (To updat weight and bias)

$$w_{3new} = w_{3\,old} - \eta \frac{\partial Loss}{\partial w_{3\,old}}$$

$$b_{2\,new} = b_{2\,old} - \eta \frac{\partial L}{\partial b_{2\,old}}$$

$$w_{1\,new} = w_{1\,old} - \eta \boxed{\frac{\partial L}{\partial w_{1\,old}}}$$

$$\frac{\partial L}{\partial w_{1\,old}} = \frac{\partial L}{\partial O_2} \times \frac{\partial O_2}{\partial O_1} \times \frac{\partial O_1}{\partial w_{1\,old}} \Bigg] \text{ chain rule}$$



$$w_{11\,new} = w_{11\,old} - \eta \frac{\partial L}{\partial w_{1\,old}}$$

$$w_{q\,new} = w_{q\,old} - \eta \boxed{\frac{\partial L}{\partial w_{q\,old}}}$$

$$\frac{\partial L}{\partial w_{q\,old}} = \frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial w_{q\,old}}$$

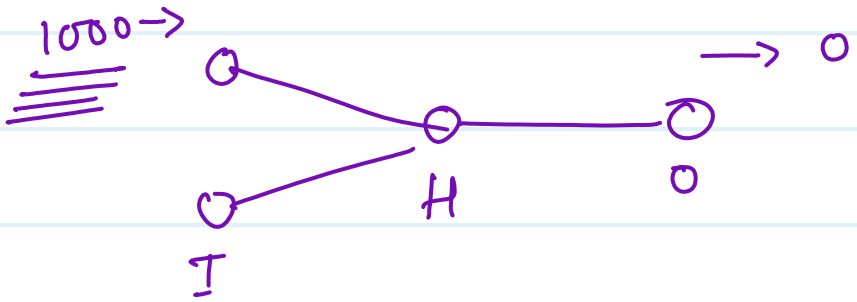
$$w_{3\,new} = w_{3\,old} - \eta \boxed{\frac{\partial L}{\partial w_{3\,old}}}$$

$$\frac{\partial L}{\partial w_{3\,old}} = \left(\frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{3\,old}}\right) +$$

$$\left(\frac{\partial L}{\partial O_3} \times \frac{\partial O_3}{\partial O_{22}} \times \frac{\partial O_{22}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{3\,old}}\right)$$

7) epoch →

Dataset – 1000



8) Batch size & iteration :-

$\frac{1000}{40} \Rightarrow 25$

epoch = 40 iteration

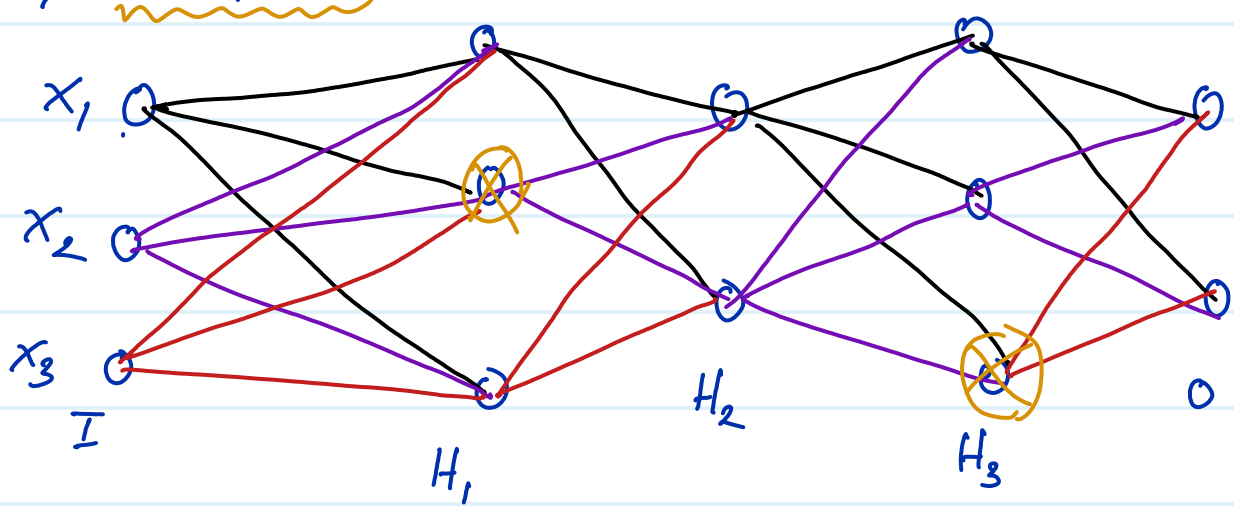
Dataset – 1000
Batch size – 25
iteration – 40

epoch = 5

total iteration = 5×40
= 200

9) Dropout :-

overfitting



10) Base equation -

$$Z = wx + b$$

$$\boxed{y = mx + c}$$

Support vector machine